Econometrics Midterm

R Functions · 5 code exhibits · W2, W5, W6 · 33 questions

Exhibit A · Lab2.R — descriptive statistics (W2)

Used in questions 1–7

```
1  practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
2  competence <- practice$COMPETENCE
3  breaks <- practice$BREAKS
4  grade <- practice$GRADE
5  hist(competence, breaks=15)
6  summary(breaks)
7  sum((competence-mean(competence))^2)/(length(competence)-1)
8  var(competence)
9  quantile(breaks)
10 IQR(breaks)
11 boxplot(grade)
12 abline(h=median(grade), col="red")
13 par(mfrow = c(1, 2))
14 boxplot(grade)
15 vioplot(grade)
```

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1. [Exhibit A] Line 1. What does read.csv() return, and what do the arguments header = TRUE and na.strings = "." do?

2. [Exhibit A] Line 5. Is breaks in hist(competence, breaks=15) the variable breaks created on line 3? What does it do?

3. [Exhibit A] Line 6. summary(breaks) returns Min. 0, 1st Qu. 1, Median 2, Mean 3.185, 3rd Qu. 4.5 and Max. 12. What does summary() report, and what does Mean > Median tell you?

4. [Exhibit A] Lines 7–8 (competence has 27 values; the sum of squared deviations is about 124.07). What does line 7 compute step by step, and why does line 8 give the same number?

5. [Exhibit A] Lines 9–10. quantile(breaks) returns 0, 1, 2, 4.5 and 12. What do these numbers represent, and what does IQR(breaks) return?

6. [Exhibit A] Lines 11–12. What does boxplot(grade) draw, and why does the red line from abline(h = median(grade)) fall exactly on the bold line in the box?

7. [Exhibit A] Lines 13–15. What does par(mfrow = c(1, 2)) do, and what appears on screen after lines 14 and 15? (vioplot() comes from the vioplot package.)

Exhibit B · Lab4.R — functions and strings (W2)

Used in questions 8–14

```
1  library(readxl)
2  library(stringr)
3  to12 <- function(str){
4    hour=as.integer(str)
5    if (hour>12) {
6      return(paste(hour-12, "PM"))
7    } else {
8      return(paste(hour, "AM"))
9    }
10 }
11 hasMultipleLoc <- function(str){
12   str_detect(str, fixed(","))
13 }
14 data=read_excel("Café.xlsx")
15 data <- as.data.frame(data)
16 colnames(data)=c("type","location","openingHour","closingHour")
17 (openingHour <- sapply(data$openingHour, to12))
18 data$openingHour=openingHour
19 (locationText <- ifelse(sapply(data$location, hasMultipleLoc), "Yes", "No"))
20 data$hasMultipleLoc=locationText
```

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8. [Exhibit B] Lines 3–10. What does to12() do, what are the roles of as.integer() and paste(), and what does to12("22") return?

9. [Exhibit B] Lines 3–10. What do to12("1") and to12("12") return, and which of the two shows a flaw in the function?

10. [Exhibit B] Lines 11–13. What does hasMultipleLoc() return for "Seoul, Suwon" and for "Seoul", and what does fixed() do?

11. [Exhibit B] Lines 11–13. hasMultipleLoc() has no return() statement. What does it return, and why?

12. [Exhibit B] Lines 14–16. What does each of the three lines do?

13. [Exhibit B] Line 17. What does sapply() do here, and why not simply call to12(data$openingHour)?

14. [Exhibit B] Lines 18–20. What does ifelse() do on line 19, and what do lines 18 and 20 change in data?

Exhibit C · regression.R (condensed) — OLS, inference and prediction (W5)

Used in questions 15–21

```
1  social <- read.csv("social.csv", header=T)
2  perform=social$cryp_scoreFINAL
3  eye=social$RMET_mean
4  skill=social$Smean_mean
5  ravens=social$ravens_mean
6  plot(perform ~ eye)
7  eye_regr=lm(perform ~ eye)
8  abline(eye_regr, col="blue")
9  abline(h=mean(perform), col="red")
10 abline(v=mean(eye), col="green")
11 skill_regr=lm(perform ~ skill)
12 skill_summary=summary(skill_regr)
13 confint(skill_regr, level=0.99)
14 (skill_coef=cor(perform,skill)^2)
15 round(skill_summary$r.squared-skill_coef,16)
16 mult_lm=lm(perform ~ skill + ravens)
17 (mult_summary=summary(mult_lm))
18 predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval="confidence",
   level=0.95)
19 points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
20 pred <- predict(eye_regr, data.frame(eye=points), interval="confidence", level=0.95)
21 polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
   border=NA)
```

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15. [Exhibit C] Lines 6–8. Why is perform written first in perform ~ eye, what does lm() return, and what does abline() draw when it is given that result?

16. [Exhibit C] Lines 9–10. What do abline(h = ...) and abline(v = ...) add, and what do they show about the blue OLS line?

17. [Exhibit C] Lines 12–13. What does summary() report for an lm model, and what does confint(skill_regr, level = 0.99) return compared with the default?

18. [Exhibit C] Lines 14–15. The R-squared of perform ~ skill is about 0.0459. What does line 14 compute, how is it related to that R-squared, and what is round(..., 16) for?

19. [Exhibit C] Lines 16–17. The summary reports coefficients 0.1712 (skill) and 0.1746 (ravens), R-squared 0.0745 and F(2, 82) = 3.3 with p = 0.04184. What does the F-test test, where do 2 and 82 come from, and what is the conclusion at 5%?

20. [Exhibit C] Line 18. What does predict() do with the data frame, and what does the result look like with interval = "confidence"?

21. [Exhibit C] Lines 19–21. What do seq(), predict() and polygon() do together?

Exhibit D · simple regression demonstration.R (condensed) — simulation of tests and intervals (W5)

Used in questions 22–27

```
1  set.seed(4201)
2  slope=0.2; intercept=3; sampleSize=10; sampleCnt=10000
3  estimator=c(); t=c(); pValue=c()
4  for(rowIdx in 1:sampleCnt) {
5    fertilizer=rnorm(sampleSize,10,2)
6    yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
7    coef=summary(lm(yield~fertilizer))$coefficients
8    estimator[rowIdx]=coef[2,1]
9    t[rowIdx]=coef[2,3]
10   pValue[rowIdx]=coef[2,4]
11 }
12 hist(estimator)
13 alpha=0.05
14 rejection=ifelse(pValue<=alpha, "Reject", "Fail to Reject")
15 (result=table(rejection))
16 barplot(result, main="Null Hypothesis Rejection")
17 sampleSize=12; slope=2; sampleCnt=20
18 results=data.frame(index=1:sampleCnt)
19 for(rowIdx in 1:sampleCnt) {
20   fertilizer=rnorm(sampleSize,10,2)
21   yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
22   fer_regr=lm(yield ~ fertilizer)
23   results[rowIdx,"slope"]=fer_regr$coefficients[2]
24   interval=confint(fer_regr)
25   lowerBound=interval["fertilizer","2.5 %"]
26   upperBound=interval["fertilizer","97.5 %"]
27   results[rowIdx,"inside"]=lowerBound<=slope && slope<=upperBound
28 }
```

22. [Exhibit D] Lines 1–6. What do set.seed() and the two rnorm() calls do, and what is the true model that generates yield?

23. [Exhibit D] Lines 7–10. What is coef, and what do coef[2,1], coef[2,3] and coef[2,4] contain?

24. [Exhibit D] Line 12. What does the histogram of estimator show, and where should it be centred?

25. [Exhibit D] Lines 13–16. What do ifelse(), table() and barplot() do, and what does the share of "Reject" estimate here?

26. [Exhibit D] If line 2 set slope=0 and the simulation were run again, what would the share of "Reject" estimate, and about what value should it be?

27. [Exhibit D] Lines 17–28. What do results[rowIdx, "slope"] and interval["fertilizer", "2.5 %"] select, when is inside TRUE, and about how many of the 20 rows should be TRUE?

Exhibit E · W6 — dummy variables and interactions (lecture examples, Wooldridge Ch.7)

Used in questions 28–33

```
1 m1 <- lm(wage ~ female + educ, data = df)
2 m2 <- lm(wage ~ educ*female, data = df)
3 m3 <- lm(wage ~ educ + as.factor(region), data = df)
4 library(car)
5 linearHypothesis(m2, c("female = 0", "educ:female = 0"))
6 restricted <- lm(wage ~ educ + exper, data = df)
7 unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
8 anova(restricted, unrestricted)
9 lpm <- lm(inlf ~ educ + age, data = df)
```

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28. [Exhibit E] Line 1. female = 1 for women and 0 for men. How is the coefficient of female interpreted?

29. [Exhibit E] Line 2. Which terms does educ*female put into the model, and what do the coefficients of educ:female and of female measure?

30. [Exhibit E] Line 3. region has several categories coded as numbers. What does as.factor() change, and what does lm() then create?

31. [Exhibit E] Lines 4–5. What does linearHypothesis() test here, which kind of test does it report, and what would rejecting the null mean?

32. [Exhibit E] Lines 6–8. What does anova(restricted, unrestricted) compute, how many restrictions are tested, and what does the test assume?

33. [Exhibit E] Line 9. inlf is 1 if a woman is in the labor force and 0 otherwise. What model does lm() estimate here, what does a coefficient measure, and what are two drawbacks?

Econometrics Midterm — Model answers

1. W2 · read.csv() and its arguments
read.csv(): reads a comma-separated file and returns a **data frame**
header: TRUE means the **first row** holds the column names, not data
na.strings: every "." in the file is read as **NA** (missing), so it does not enter calculations as a value
Key results: data frame · first row · NA

2. W2 · The breaks argument of hist()
No: breaks= here is an **argument** of hist(), matched by its name, not the variable breaks
It sets the **number of bins** (bars) of the histogram, here about 15
A single number is only a **suggestion**: R chooses nearby "pretty" break points, so the exact count can differ
Key results: argument · number of bins · suggestion

3. W2 · Reading summary()
summary(): for a numeric vector reports the minimum, the first **quartile**, the median, the mean, the third quartile and the maximum
Here the mean (3.185) is above the median (2): a few **large values**, up to 12, pull the mean up
The distribution is **positively (right) skewed**
Key results: quartile · right · large values

4. W2 · Sample variance written out
(competence - mean(competence))^2 squares each deviation from the mean, and sum() adds them (about 124.07)
length(): gives n = 27, and dividing by **n - 1** = 26 gives about **4.77**
var(): computes the same **sample variance**, with the same n - 1 denominator, so line 8 returns the same value
Key results: sample variance · n - 1 · 4.77 · var

5. W2 · quantile() and IQR()
quantile(): by default returns the 0%, 25%, 50%, 75% and 100% points: the **minimum**, the first **quartile**, the median, the third quartile and the maximum
IQR(): the **third quartile minus** the first, 4.5 - 1 = **3.5**, the spread of the middle 50% of the students
Unlike the standard deviation, it is not pulled by the large values in the right tail
Key results: quartile · minimum · 3.5

6. W2 · boxplot() and abline(h =)
boxplot(): draws a **box** from the lower to the upper hinge (about Q1 to Q3) with a bold line at the median
The **whiskers** reach the most extreme points within 1.5 times the IQR, and points beyond them are drawn as outliers
abline(): with h set to a value draws a **horizontal** line there
The bold line is the **median**, so the red line at median(grade) falls on it
Key results: box · whiskers · median · horizontal

7. W2 · par(mfrow =) and vioplot()
par(): sets graphics parameters, and mfrow = c(1, 2) splits the window into **1 row and 2 columns**
So the boxplot (line 14) and the violin plot (line 15) of grade appear **side by side**
vioplot(): draws a smoothed **density** mirrored on both sides, with a small box plot inside
The layout stays in effect **until** it is reset, for example with par(mfrow = c(1, 1))
Key results: 1 row and 2 columns · side by side · density · until

8. W2 · What to12() does
to12(): converts an hour on the **24-hour** clock, given as text, into a 12-hour label
as.integer(): turns the text into a **number**, so it can be compared with 12 and reduced by 12
paste(): **joins** its arguments with a space
to12("22"): 22 > 12, so it returns paste(10, "PM"), that is "**10 PM**"
Key results: 24-hour · number · joins · 10 PM

9. W2 · Testing to12() at the edges
to12("1"): 1 is not greater than 12, so it returns **"1 AM"**, which is correct
to12("12"): 12 is not **greater than** 12 either, so it returns **"12 AM"**
That is the flaw: 12 o'clock in 24-hour time is **noon**, 12 PM
The condition hour > 12 treats 12 as morning, so 12 needs its own case (and 0, midnight, would give "0 AM")
Key results: 1 AM · 12 AM · noon · > 12

10. W2 · str_detect() and fixed()
str_detect(): returns **TRUE** or FALSE depending on whether the string **contains** the pattern
So hasMultipleLoc("Seoul, Suwon") is TRUE and hasMultipleLoc("Seoul") is FALSE
fixed(): makes "," a **literal** string matched character by character, not a regular expression
Key results: TRUE · contains · literal

11. W2 · A function without return()
An R function returns the value of the **last** expression it evaluates, even without return()
The last (and only) expression is str_detect(...), so its TRUE or FALSE is returned **automatically**
return() is needed mainly to leave a function early, as in the two branches of to12()
Key results: last · automatically · TRUE

12. W2 · read_excel(), as.data.frame() and colnames()
read_excel(): from readxl (line 1), reads the **Excel** file Café.xlsx into a **tibble**
as.data.frame(): converts the tibble into an ordinary base R data frame
colnames(data) = c(...) **renames** the four columns to type, location, openingHour and closingHour
Key results: Excel · tibble · rename

13. W2 · Why sapply()
sapply(): calls to12() on **each element** of the column and simplifies the results into a character **vector**
to12() uses **if**, which needs a single TRUE or FALSE
Given the whole column at once, the condition has many values and if() stops with an error (in current versions of R)
Key results: each element · vector · if

14. W2 · ifelse() and the new columns
ifelse(): checks a **condition** for **each element** and returns "Yes" where it is TRUE and "No" where it is FALSE
Line 18 **overwrites** the openingHour column with the 12-hour labels made on line 17
Line 20 adds a **new column**, hasMultipleLoc, with the Yes or No for each café
Key results: condition · each element · Yes · overwrites

15. W5 · plot(), lm() and abline(model)
perform ~ eye: the variable left of ~ is the **dependent** (outcome) variable, so plot() puts perform on the y-axis and eye on the x-axis
lm(): fits a **linear model** by OLS and returns a model object holding the intercept and the slope
abline(): given that object, draws the fitted **regression line** in blue
Key results: dependent · linear model · regression line

16. W5 · Lines at the means
abline(): with h draws a **horizontal** line at the **mean of perform**, and with v a **vertical** line at the mean of eye
The two lines cross at the point of means
The OLS line always **passes through** that point, because the OLS intercept equals mean(perform) - slope × mean(eye)
Key results: horizontal · vertical · mean of perform · passes through

17. W5 · summary() and confint(level = 0.99)
summary(): reports the **coefficient table** (Estimate, Std. Error, t value and p-value of each term), the residual standard error, the **R-squared** and adjusted R-squared, and the F-statistic
confint(): returns a **confidence interval** for each coefficient, by default at the 95% level
level: 0.99 gives 99% intervals (columns 0.5 % and 99.5 %), which are **wider**
Key results: coefficient table · R-squared · confidence interval · wider

18. W5 · cor() squared and round(..., 16)
cor(): computes the Pearson **correlation** of perform and skill, and ^2 squares it
In a simple regression the squared correlation **equals** the R-squared, so line 14 also gives about 0.0459
round(): rounds the difference to 16 decimals so that tiny **floating**-point differences between the two calculations are not read as a real gap
Key results: correlation · equals · floating

19. W5 · The multiple regression and its F-test
F-test: H_0 says the skill and ravens slopes are **both** 0, H_1 that at least one is not 0
2: the number of **restrictions**, one per predictor tested
82: the residual degrees of freedom, **n - k - 1** = 85 - 2 - 1
Since **p** = 0.04184 < 0.05 we **reject** H_0: the model explains perform better than a model with no predictors
Key results: both · restrictions · n - k - 1 · reject

20. W5 · predict() with interval = "confidence"
predict(): plugs the **new data**, two cases with the same variable names as the model, into the fitted equation and returns the **predicted** values (about 0.3475 and 0.3441)
interval: "confidence" adds the **lower and upper** bounds, so the result is a matrix with the columns fit, lwr and upr, one row per case
The interval is for the average perform at those predictor values, not for one new group
Key results: predicted · new data · lwr · 0.3475

21. W5 · Drawing the confidence band
seq(): creates 10,000 **evenly spaced** values of eye from min(eye) - 3 to max(eye) + 3
predict(): gives the fitted value and the 95% **lower and upper** bounds at each of those points (columns 2 and 3 of pred)
polygon(): joins the upper bounds (reversed) and the lower bounds into one closed shape and **fills** it in transparent grey, the **shaded** confidence band around the line
Key results: evenly spaced · lower and upper · fills

22. W5 · set.seed() and rnorm()
set.seed(): fixes the start of the random number generator, so every run gives the same, **reproducible** draws
rnorm(sampleSize, 10, 2) draws 10 fertilizer values from a **normal** distribution with mean 10 and standard deviation 2
rnorm(sampleSize, 0, 1) draws the error term u with mean 0 and standard deviation 1
So the true model is yield = 3 + 0.2 × fertilizer + u
Key results: reproducible · normal · 3 + 0.2

23. W5 · The coefficient matrix
summary(...)$coefficients: is the coefficient **matrix**, one row per term ((Intercept) and fertilizer) with the columns Estimate, Std. Error, t value and p-value
coef[2,1] is the slope **estimate** of fertilizer, coef[2,3] its **t value** and coef[2,4] its **p-value** for the test of slope = 0
Each is stored at position rowIdx, so after the loop each vector holds 10,000 values
Key results: matrix · estimate · t value · p-value

24. W5 · hist(estimator)
hist(): of the 10,000 stored slopes shows the **sampling distribution** of the OLS slope estimator
It should be centred near the **true slope** 0.2, because OLS is **unbiased** under these assumptions
Its spread is the sampling variability that the standard error measures
Key results: sampling distribution · 0.2 · unbiased

25. W5 · Counting rejections
ifelse(): gives "Reject" where pValue <= 0.05 and "Fail to Reject" otherwise, one **label** per sample
table(): **counts** how many samples carry each label
barplot(): draws one **bar** per label
The true slope is 0.2, not 0, so the null is false and the share of Reject estimates the test's **power**
Key results: labels · counts · bar · power

26. W5 · If the true slope were 0
With slope=0 the null hypothesis is **true**, so every rejection is a wrong one
The share of Reject then estimates the **Type I** error rate
It should be close to alpha, about **0.05** (5%), apart from simulation noise
Key results: Type I · 0.05 · null is true

27. W5 · Filling results and checking coverage
results[rowIdx, "slope"] = ... writes one **cell**, chosen by row number and column name, and creates the column the first time
interval["fertilizer", "2.5 %"] picks the **lower** bound in the fertilizer row of the confint() matrix ("97.5 %" is the upper bound)
&& is TRUE only if **both** comparisons hold, so inside is TRUE when the true slope 2 lies within the interval
About **95%** of such intervals cover the true slope, roughly 19 of 20, though one run can differ by chance
Key results: cell · lower · both · 19

28. W6 · A dummy variable in lm()
The coefficient of female is the **difference** in expected wage between women and the **base group**, men
It is measured **holding educ** constant, that is, for a man and a woman with the same education
Graphically it shifts the **intercept**: both groups have the same slope on educ
Key results: difference · base group · holding educ · intercept

29. W6 · The * operator and the interaction term
educ*female: expands to educ + female + educ:female, the two **main effects** plus their **interaction**
The educ:female coefficient is the difference in the **slope** of educ between women and men (the difference in the return to education)
female: the difference in the **intercept**, the wage gap at educ = 0
Key results: main effects · interaction · slope · intercept

30. W6 · as.factor() with several categories
as.factor(): treats region as a **categorical** variable, not as one number with a single slope
lm(): creates one **dummy** for each region except one, the **base** (reference) category
Leaving one out avoids the dummy variable trap (perfect collinearity with the intercept)
Each dummy coefficient is the wage difference from the base region, holding educ constant
Key results: categorical · dummy · base · dummy variable trap

31. W6 · linearHypothesis() on m2
linearHypothesis(): from the **car** package, tests the restrictions female = 0 and educ:female = 0 **jointly**
For an lm model it reports an **F-test**, here with 2 restrictions
The null says women and men have the **same intercept** and the same slope, one line for both
Rejecting it means the wage–education relationship differs by gender
Key results: jointly · F-test · car · same intercept

32. W6 · anova() and the Chow test
anova(): with two **nested** models **compares** their residual sums of squares with an **F-test**
The unrestricted model adds female, educ:female and exper:female, so **3** restrictions are tested
This is the **Chow** test of whether one regression fits both women and men
It assumes the **same error variance** in both groups
Key results: F-test · nested · 3 · Chow · same error variance

33. W6 · lm() with a binary dependent variable
lm(): with a 0/1 dependent variable estimates a **linear probability model**
A coefficient is the change in the **probability** that inlf = 1 for a one-unit increase in the regressor, holding the others fixed
Drawback 1: fitted values can fall **outside 0 and 1**
Drawback 2: the error variance depends on x, so the model is **heteroskedastic** and the usual standard errors need care
Key results: linear probability model · probability · outside 0 and 1 · heteroskedastic